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Soft-Output Deep LAS Detection for Coded MIMO Systems: A Learning-Aided LLR Approximation

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Abstract—The multiple-input-multiple-output-orthogonal frequency division multiplexing (MIMO-OFDM) receiver aims to softly decode the transmitted information from the observed received signal. However, soft-output detection requires additional computation and leads to higher overall detection complexity in high-dimensional MIMO and higher-order modulation. Therefore, accurate and low-complexity soft-output detection is a challenging task in such systems. The conventional likelihood ascent search (LAS) detectors perform well in large antenna setups, however, multiple symbol updates and the soft-output estimation using brute force search further boost its complexity. In this paper, we propose a model-based soft-output LAS detector to jointly detect and precisely estimate the soft output in MIMO-OFDM systems. Furthermore, we propose a data-driven deep-LAS (Deep LAS) architecture for MIMO detection which is a multi-layer perception (MLP), and a gated recurrent unit (GRU)-based hybrid trainable learning framework to unfold the proposed two-stage LAS algorithm by directly learning the soft output from the received equalized signals. Numerical results demonstrate that the proposed two-stage soft-output LAS detector precisely computes the log-likelihood ratio (LLR) and provides better performance than the conventional LAS detector. Alternatively, the proposed Deep LAS efficiently estimates the LLR values by achieving a performance gain of 2.55 dB and 3 dB compared to the conventional LAS algorithm. Furthermore, the proposed Deep LAS outperforms the counterpart model-based and standalone data-driven learning schemes and provides a comparable signal-to-noise ratio (SNR) gap of 0.4 dB and 1.2 dB with the optimal soft output sphere decoding (SD) to achieve a BER of 10^{-5} for 4-QAM and 16-QAM, respectively.

Index Terms—Deep-likelihood ascent search (Deep LAS), LLR approximation, machine learning, MIMO-OFDM, soft-output MIMO detection.

I. INTRODUCTION

Multiple-input-multiple-output (MIMO) is one of the key technologies to meet the constantly growing throughput requirement in modern wireless communication systems. MIMO systems provide a spatial degree of freedom through multi-antenna transmission between the transmitter and receiver, enabling spatial multiplexing, transmit diversity, and multi-user transmission to enhance the overall system throughput and ensure higher spectral efficiency compared to single-

input-single-output (SISO) systems [1], [2]. However, multiple data stream transmissions through multiple antennas ensure spatial multiplexing gain but it results in inter-stream interference at the receiver. On the other hand, orthogonal frequency division multiplexing (OFDM) is a promising technique to effectively combat the effect of frequency-selective fading through simplified transceiver architecture that utilizes multiple narrow-band orthogonal channels instead of wide-band channels. Combining MIMO with OFDM (MIMO-OFDM) leads to improved systems capacity due to the utilization of spatial diversity, time coding, and the inter-symbol interference cancellation [3]. However, accurate symbol detection at the receiving end is imperative to fulfill the potential of MIMO-OFDM systems which poses design aspect challenges including low complexity near-optimal receiver. To this end, the received signal in MIMO systems is either detected coherently or non-coherently [4]. Coherent detection requires explicit channel state information (CSI) at the receiver end while non-coherent detection either avoids or implicitly performs channel estimation [5].

The main objective of MIMO-OFDM receivers is to restore the originally transmitted information bits from the received noisy signals with less possible computational complexity. The complexity of the MIMO receivers increases exponentially with the transmit-receive chains and the modulation order making it a non-deterministic polynomial (NP) hard problem [6]. In this regard, the optimal detectors achieve better performance at the cost of higher computational complexity, while the sub-optimal detection techniques may incur performance degradation. Moreover, generating soft output for the detected information bits is equally important to further enhance the performance of MIMO-OFDM receivers. Therefore, a low-complexity soft-output detector is required that can achieve bit-error-rate (BER) performance nearly equal to the optimal detectors.

A. Related work and objective

Several model-based optimal and sub-optimal MIMO detection algorithms have been proposed in the literature, which are broadly characterized into maximum likelihood detectors [7]–[10], linear detectors [11], [12], interference cancellation based algorithms [13], graph-based algorithms [14], sparsity-boosted algorithms [15], lattice reduction-based algorithms [16], and neighborhood search-based algorithms [17]–[22]. The maximum likelihood and maximum a posteriori (MAP) receivers simultaneously recover all the symbols and achieve

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superior BER performance with polynomial order computational complexity, which makes them infeasible for real-time implementation [7]. A variant of optimal detectors, sphere decoding (SD) is proposed in [8] which controls the complexity of the optimal detection to some extent by limiting the search space to a fixed circular region. The soft-output SD algorithms for distributed antenna and MIMO-OFDM systems are proposed in [9] and [10], respectively, where the authors demonstrate that for worst case scenario, the performance of SD can be further improved by precisely estimating the soft-output information. However, the detection complexity of SD is still higher in high-dimensional MIMO systems. On the other hand, linear detectors such as zero-forcing (ZF) [11] and minimum mean square error (MMSE) [12] have lower detection complexity but incur higher performance loss compared to the optimal detection schemes. Interference cancellation-based detection algorithms such as successive interference cancellation (SIC) and parallel interference cancellation (PIC) are proposed which follow layer-wise detection of received symbols by recovering a particular layer of symbols while treating the rest as an interference. However, these detection algorithms also exhibit a performance gap with the optimal detectors [13].

Neighborhood search algorithms (NSAs) have been extensively explored in the literature to address the limitations of other detection algorithms. NSA augmented with the sparsity and lattice reduction (LR) techniques are studied in [23] and [24], respectively. The likelihood ascent search (LAS) algorithm is used for MIMO detection which performs significantly well in higher antenna setups. The LAS utilizes a linear equalized signal as an initial guess and generates a search space for the initial symbol vector based on the symbol update rule. The best solution vector from the search space is selected iteratively based on the likelihood cost [17]. LAS algorithms are classified into one-stage LAS (1-LAS) and multi-stage LAS (M-LAS) based on the number of simultaneous symbol updates per iteration [19]. M-LAS provides better BER performance compared to 1-LAS as it searches for more neighbors with multiple symbol updates, but has significantly higher computational complexity than 1-LAS. The authors in [20] propose a random list-based LAS algorithm for large-scale uplink multi-user MIMO systems. Similarly, LAS is used in conjunction with the hybrid detectors to improve the performance of MIMO receivers with imperfect CSI [21] while the performance of unconstrained neighborhood search-based LAS algorithm is analyzed in [22]. All the variants of the model-based LAS algorithm mentioned earlier either employ hard output detection or use brute force techniques for log-likelihood ratio (LLR) approximation to generate the soft output. However, brute force-based LLR approximation improves BER performance but puts an additional computation burden on the receiver in the case of higher-order modulation and high-dimensional MIMO systems.

As an alternative to the model-based detection schemes, the data-driven detectors utilize emerging machine learning (ML) tools to recover the desired information. Due to their strong learning capability, ML has been utilized in different research areas, such as computer vision, image processing,

and language processing, and has recently attracted the interests of the research community to explore new aspects and solve different problems in wireless communication systems [25]–[28]. To utilize ML in modeling the physical layer for contemporary communication systems, researchers have extensively explored deep neural network (DNN) for channel estimation; however, there exist several works which utilize it for solving MIMO detection problem [29]–[37]. Unlike model-based detectors, ML detectors are trained to learn the channel behavior and directly map the channel-corrupted received data into the actual transmitted data in a data-driven fashion without relying on specific detection rules. Once properly trained, the data-driven MIMO-OFDM receivers can implement the complicated and complex MIMO detection problem while maintaining reasonable computational complexity.

ML-based MIMO receivers can be classified into two main categories: conventional deep neural network (DNN)-based receivers and deep-unfolded networks. The former uses the DNN architecture while the latter uses the learning networks inspired from the model-based algorithms. The authors in [29] utilize DNN to softly demodulate in conventional SISO systems. In [30], the authors propose a blind detection framework based on supervised learning to detect the transmitted symbols in the MIMO system with no explicit knowledge of the channel at the receiver. The authors in [31] employed sequential learning-aided deep-SIC for multi-user MIMO systems to jointly detect the received symbols in a data-driven manner. Similarly, the authors in [32] and [38] used a DNN i.e., MLP and convolutional neural network (CNN) to improve the performance of the LAS algorithm in a multi-user MIMO system, where the received signal is first detected using the LAS algorithm and then passed through a trained DNN with optimal detection target data. However, it requires CSI at the receiver and incurs extra training overhead. Considering the dependence of computational complexity on the number of neurons and hidden layers, the authors in [33] proposed a partial DNN-based detection scheme for massive MIMO systems. Specific DNN architectures named DetNet and MMNet are proposed for MIMO detection in [35], [39] and [36], respectively. The former work is based on the iterative unfolding of gradient descent algorithm while the latter uses an iterative soft-thresholding algorithm to leverage temporal and spectral correlation which have lower computational complexity and improved symbol-error rate compared to the linear detectors. Unlike [35] and [36], which require perfect CSI and a higher number of antennae at the transmitter than the receiver, the authors in [37] proposed a model-driven DNN framework for joint channel estimation and signal detection in MIMO systems. However, the proposed detection scheme exhibits a high SNR gap compared to the optimal MAP and SD detectors considering the M-QAM constellations. Similarly, the performance of model-driven linear detectors for acoustic OFDM systems is studied in [40]. Other learning methods such as CNN [41], long short-term memory (LSTM) [42], bi-directional long short-term memory (Bi-LSTM) [43], and gated recurrent unit (GRU) [44] are used for end-to-end learning in communication systems. The authors in [45] proposed a CNN-based DemodNet to approximate the soft output and

achieve the BER performance close to the exact LLR approximation in SISO systems with AWGN channel. In contrast, the authors in [46] used a twin neural network structure based on multiple DNN to detect real-valued multi-level BPSK symbols. A learning-based SD algorithm is proposed in [47] and [48] for higher order modulation where DNN is utilized to approximate the sphere radius and initial candidate for SD in MIMO systems, respectively. Similarly, a learning-based iterative search algorithm (LISA) is proposed in [49] to softly detect over the fixed and varying MIMO channel. These detectors perform well in fixed MIMO channels however increase in the number of transmit-receive chains leads to a higher SNR gap with optimal detectors.

The aforementioned literature on data-driven approaches either propose hard output detectors [30], [32], [33], [35]–[37], [39], [46], [47] for MIMO systems or consider simple SISO channels [29], [45]. These approaches detect symbols directly from the received data in modulation-independent scenarios and do not consider the impact of other OFDM blocks while it is more desirable to follow the block-wise optimization of communication systems in a data-driven fashion. Moreover, the performance of these detectors degrades with the increasing number of antennas. To fill the above gaps, the objectives of this paper are to: 1) propose a new computationally efficient model-based soft-output detector and 2) design an efficient data-driven soft-output detector by making use of a robust DNN model to softly decode the transmitted information and avoid computationally inefficient brute-force technique for LLR approximation in MIMO-OFDM systems.

B. Contribution and paper organization

This paper proposes a computationally efficient two-step soft-output LAS and a hybrid DNN-based Deep LAS detectors for MIMO-OFDM systems which learn the LLR values from the pre-processed received symbol data. The contributions of this paper are multi-fold which are summarized as follows.

- We propose a model-based low-complexity soft-output LAS detector for MIMO-OFDM systems. The proposed detector comprises two steps: the first step follows the detection of symbols using 1-LAS and the second step utilizes a modified-LAS algorithm to generate the soft output. The modified LAS algorithm is an updated variant of LAS used in the detection stage, which does not need the equalization step and searches for the best counter-hypothesis for the detected hypothesis within the constellation grid. The modified LAS algorithm efficiently approximates the LLR values with fewer computations involved compared to the brute-force techniques.
- To estimate the LLR values directly from the equalized receive signal, we replace the two steps in the model-based LAS with a DNN framework. In this regard, we propose a learning architecture with two DNN blocks, i.e., multi-layer perceptron (MLP) and GRU block for soft-output Deep LAS detection. The proposed MLP-GRU-based Deep LAS is inspired from the hybrid DNN models which utilize multiple trainable DNN blocks to efficiently learn the non-linearity in the input data. The

Deep LAS learns the LLR values in a data-driven fashion where the received equalized data are fed into the initial block for rough approximation and the succeeding block further refines it using extended data inputs. To the best of our knowledge, the proposed Deep LAS is not reported in the literature.

- Based on extensive simulations, we identify that the proposed model-based soft-output LAS algorithm outperforms its conventional counterparts in terms of BER. We demonstrate that the proposed MLP-GRU-based Deep LAS accurately learns the LLR values from the received preprocessed signals providing better BER performance compared to other model-based as well as standalone data-driven approaches. Furthermore, the proposed Deep LAS provides BER performance nearly equal to the optimal SD receivers.

The rest of the paper is organized as follows. Section II provides a detailed description of the MIMO-OFDM model followed by a brief overview of the MIMO detection and the computation of LLR values. Section III describes the proposed model-based two-stage LAS for efficient estimation of soft output. Section IV presents the data pre-processing steps followed by the DNN architecture for the proposed Deep LAS detector. Section V presents the performance evaluation of the proposed MIMO detectors. Finally, Section VI concludes the paper by summarizing the key outcomes and contributions of the paper.

C. Notations

The notations convention used in the paper are as follows. The lowercase letters denote scalars, bold-face lowercase letters denote vectors, and bold-face uppercase letters denote matrices. The transpose and conjugate transpose operations are denoted by $(\cdot)^T$ and $(\cdot)^H$, respectively. The n th value of $\mathbf{x}$ in the m th iteration is denoted by $\mathbf{x}_n^{(m)}$, while the identity matrix, real and imaginary parts of complex-valued vectors are denoted by $\mathbf{I}$, $\Re(\cdot)$, and $\Im(\cdot)$, respectively.

II. SYSTEM MODEL

The considered communication scenario involves MIMO-OFDM systems with Vertical Bell Laboratories Layered Space-Time (V-BLAST)-based spatially multiplexed transmissions, where the transmitter is equipped with N_t number of transmit antennas and the receiver is equipped with N_r number of antennas for data reception as shown in Fig. 1. The binary information bits are encoded, interleaved, and mapped into modulated symbols using a multilevel quadrature amplitude modulation (M-QAM) scheme at the transmitter. For OFDM transmission, the transmitter performs inverse fast Fourier transform (IFFT) and inserts a cyclic prefix (CP) into the M-QAM modulated complex symbols before transmitting it over the wireless channel. The transmitted complex signal vector over a fixed symbol time is given as $\mathbf{x} = [x_1, \dots, x_{N_t}]^T \in \mathbb{C}^{N_t \times 1}$. Let us assume that $x_i \in \mathcal{A}$ is the symbol transmitted over i th antenna, where $\mathcal{A} = \mathbb{A} + j\mathbb{A}$ denote the finite set of symbols constellation from M-QAM modulation with $\mathbb{A} = \left\{ -\left(\sqrt{M}-1\right), \dots, -3, -1, 1, 3, \dots, \left(\sqrt{M}-1\right) \right\}$.

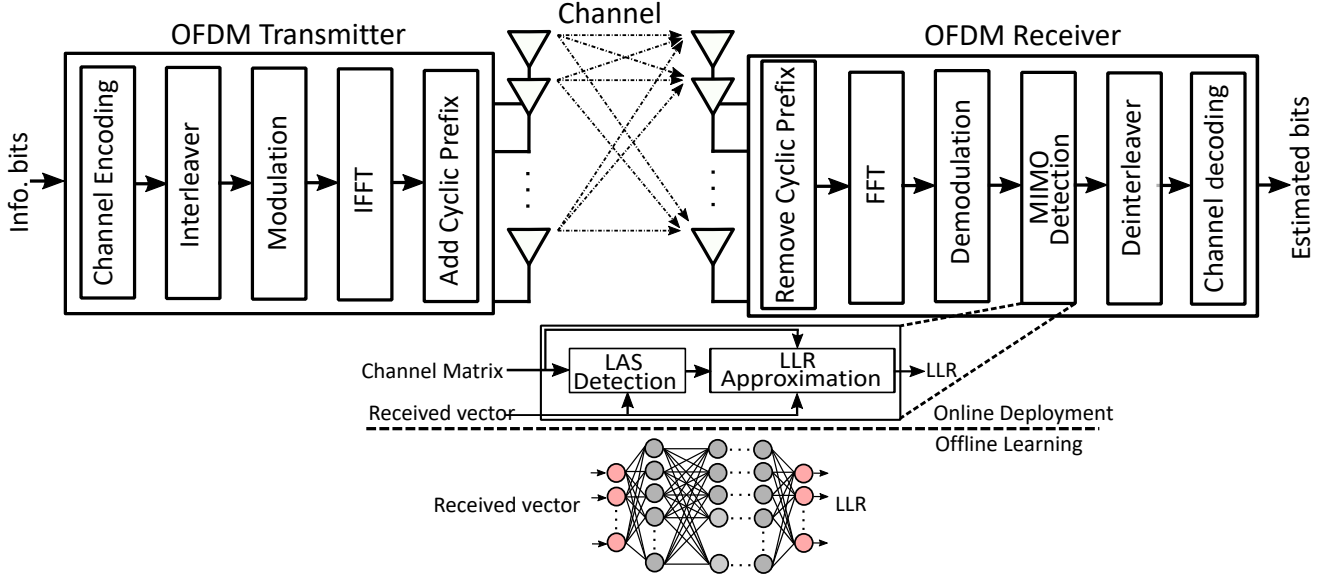


Fig. 1: MIMO transceiver model with an illustration of model-based and data-driven symbols detection at the receiver.

The complex M-QAM transmit symbols are demultiplexed into N_t number of independent data streams. Omitting the sub-carrier index for notational brevity, the channel output based on the generalized MIMO channel model over a unit frame time can be expressed as

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (1)$$

where the vector $\mathbf{y} = [y_1, \dots, y_{N_r}]^T \in \mathbb{C}^{N_r \times 1}$ denotes the received symbol vector over a fixed symbol time, $\mathbf{n} = [n_1, \dots, n_{N_r}]^T \in \mathbb{C}^{N_r \times 1}$ is the complex additive white Gaussian noise (AWGN) with zero mean and variance σ^2 , i.e., $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_{N_r})$.

The $N_r \times N_t$ channel matrix between the transmitter and the receiver is given by

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1j} \\ h_{21} & h_{22} & \cdots & h_{2j} \\ \vdots & \vdots & \ddots & \vdots \\ h_{i1} & h_{i2} & \cdots & h_{ij} \end{bmatrix}, \quad (2)$$

where h_{ij} is the prior known channel gain between the j th transmit antenna and the i th receive antenna. The channel matrix $\mathbf{H}$ is a complex-valued Rayleigh fading channel whose coefficients follow independent identically distributed (i.i.d.) complex Gaussian distribution with zero mean and unit variance [4], [38]. The receiver removes the cyclic prefix from the received signal, passes it through the FFT block, and feeds it into the soft-output MIMO detector. The computed LLR values are deinterleaved and fed into the turbo decoder to recover the originally transmitted information bits.

A. MIMO detection and computation of LLR

The main goal of MIMO detectors is to estimate the transmitted information bit from the channel corrupted received

signal vector $\mathbf{y}$. The MIMO detection is either hard or soft-output, where the latter involves two main steps: 1) the detection of actual transmitted symbols from the received symbols, and 2) the computation of bit-wise LLR values. Based on the optimal detection technique, the MIMO detection problem is equivalent to the minimization of the Euclidean distance and is numerically expressed as

$$\hat{\mathbf{x}} = \min_{\mathbf{x} \in \mathcal{A}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2. \quad (3)$$

The optimal maximum likelihood receiver searches for the optimal solution within the search space with all the possible solution sets using the brute force search based on (3). To further facilitate an improved decoding, the $N_r \times 1$ estimated symbol vector $\hat{\mathbf{x}}$ with $\mathbb{B} = \log_2(M)$ number of bits per symbols are mapped into $N_t \times \mathbb{B}$ LLR matrix, $\mathbf{L} = [\mathbf{l}_1, \dots, \mathbf{l}_{N_t}]^T$, where $\mathbf{l}_j \forall j \in \{1, 2, \dots, N_t\}$ denotes the $\mathbb{B} \times 1$ LLR vector for the j th symbol. The LLR value of the i th bit in the j th symbol is represented by l_{ij} which corresponds to the reliability of the estimated binary coded bits. Let us denote the i th bit in the j th detected symbol by $c_{ij} \in \{0, 1\}$, then based on the logarithm of maximum a posteriori probability (log-MAP), the LLR value of i th bit in the j th symbol is given by [50]

$$l_{ij} \triangleq \log \left(\frac{\Pr(c_{ij} = 0 | \mathbf{y})}{\Pr(c_{ij} = 1 | \mathbf{y})} \right) \stackrel{(a)}{\approx} \log \frac{\sum_{\hat{\mathbf{x}} \in \mathcal{A}_i^0} \exp \left(-\frac{\|\mathbf{y} - \mathbf{H}\hat{\mathbf{x}}\|_2^2}{\sigma^2} \right)}{\sum_{\hat{\mathbf{x}} \in \mathcal{A}_i^1} \exp \left(-\frac{\|\mathbf{y} - \mathbf{H}\hat{\mathbf{x}}\|_2^2}{\sigma^2} \right)},$$

$$\stackrel{(b)}{\approx} \frac{1}{\sigma^2} \left(\underbrace{\min_{\hat{\mathbf{x}} \in \mathcal{A}_i^1} \|\mathbf{y} - \mathbf{H}\hat{\mathbf{x}}\|_2^2}_{\mathcal{F}_0} - \underbrace{\min_{\hat{\mathbf{x}} \in \mathcal{A}_i^0} \|\mathbf{y} - \mathbf{H}\hat{\mathbf{x}}\|_2^2}_{\mathcal{F}_1} \right) \quad (4)$$

where $\mathcal{A}_i^{\mathcal{U}} \in \mathcal{A} \forall \mathcal{U} \in \{0, 1\}$ denotes a symbol constellation

with uth bit of interest, (a) follows the exact computation of log-MAP approximation and (b) follows from the maximum logarithmic approximation i.e., $\log(\sum_t \exp(-x_t^2)) \approx \max_t(-x_t^2)$. The max-log-MAP approximation in (b) is computationally more efficient than (a) because it eliminates the need to evaluate complex exponential and logarithmic functions. In the MIMO-OFDM system, the max-log-MAP approximation relies on a brute force search technique with the computational complexity of $O(2^{MN_t})$ [29]. Hence, obtaining soft output using an exhaustive search increases the computational burden on the MIMO-OFDM receiver as M and N_t increase. To address this issue, we propose a two-step soft-output LAS detection algorithm for the MIMO-OFDM systems which avoids the exhaustive brute force approximation.

III. SOFT-OUTPUT LAS DETECTION

In this section, we present the proposed two-step soft-output LAS algorithm for MIMO-OFDM. The first step follows the detection of the received signal and the second step computes the counter estimate for the detected symbols to facilitate the bit-wise computation of LLR values. To address the soft-output detection problem in the aforementioned system model, the computation of LLR in (4) requires the estimation of two minima, i.e., $\mathcal{F}_0$ and $\mathcal{F}_1$. The former represents the hypothesis symbols detection while the latter denotes the corresponding counter symbols estimate. The proposed algorithm finds the LLR values in (4) by estimating the hypothesis $\mathcal{F}_0$ using the conventional LAS algorithm which is based on the minimization of the likelihood function in an iterative fashion. Unlike the brute force search, the corresponding counter estimate $\mathcal{F}_1$ is approximated using the modified LAS (mod-LAS) algorithm. The steps followed in the soft-output LAS detection are as follows.

1) **Step 1:** To calculate $\mathcal{F}_0$, the algorithm begins with an initial estimate. To obtain an initial estimate of the transmitted signal vector, denoted by $\hat{\mathbf{x}}$, the received signal vector $\mathbf{y}$ is equalized by passing through linear detectors such as ZF and MMSE. The equalized output symbols vector estimated by ZF and MMSE detector is given as [51]

$$\hat{\mathbf{x}}_{ZF} = \mathbb{Q}\left((\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y}\right) \quad (5)$$

and

$$\hat{\mathbf{x}}_{MMSE} = \mathbb{Q}\left(\left(\mathbf{H}^H \mathbf{H} + \frac{\sigma^2}{E_s} \mathbf{I}\right)^{-1} \mathbf{H}^H \mathbf{y}\right), \quad (6)$$

respectively, where $\mathbb{Q}(\cdot)$ is the hard decision operator relative to symbol alphabets in $\mathcal{A}$ and E_s denotes the symbol energy.

The LAS algorithm generates the search space for the initial estimate using the symbol update policy and searches for the best solution in the neighborhood. All the neighboring vectors are searched by the LAS algorithm iteratively and minimizes the likelihood function which monotonically decreases from one iteration to the next. The algorithm stops searching when there is no longer a chance for a decrease in likelihood value. The LAS with a single symbol update is referred to as 1-LAS, which can be extended to multi-LAS (M-LAS) with multiple

symbol updates at a time to generate the neighboring vectors. The M-LAS algorithm achieves a marginal performance gain compared to 1-LAS while the complexity increases significantly¹. Let us denote the initial solution vector obtained from the linear detector as $\hat{\mathbf{x}}^{(0)} = \hat{\mathbf{x}}_i$, where $i \in \{\text{ZF, MMSE}\}$. The likelihood function in the m th iteration can be obtained by simplifying the likelihood metric in (3) and is given as

$$\Lambda^{(m)} = \hat{\mathbf{x}}^{(m)T} (\mathbf{H}^H \mathbf{H}) \hat{\mathbf{x}}^{(m)} - 2\mathbf{y}^T \mathbf{H} \hat{\mathbf{x}}^{(m)}. \quad (7)$$

The n th entry of $\hat{\mathbf{x}}^{(m)}$ is updated in $(m+1)$ th iteration to generate the neighboring vector during the local search. The corresponding vector with one symbol updated is given by

$$\hat{x}_n^{(m+1)} = \hat{x}_n^{(m)} + \lambda_n^{(m)} \text{sign}(z_n^{(m)}), \quad (8)$$

where $\hat{x}_n^{(m)}$ and $z_n^{(m)}$ are the m th entry of $\hat{\mathbf{x}}^{(0)}$ and $\mathbf{z}^{(0)} = (\mathbf{y} - \mathbf{H}\hat{\mathbf{x}}^{(0)}) \mathbf{H}^H$, respectively. Let the n th diagonal entry of $\mathbf{Q} = \mathbf{H}^H \mathbf{H}$ is denoted by q_{nn} , then $\lambda_n^{(m)}$ can be expressed as

$$\lambda_n^{(m)} = 2 \left\lfloor \left| z_n^{(m)} \right| / (2q_{nn}) \right\rfloor, \quad (9)$$

where $\lfloor \cdot \rfloor$ operator denotes the rounding operation. The updated vectors are properly checked in each iteration whether its n th entry in $(m+1)$ th iteration, $\hat{x}_n^{(m+1)}$, lies on the QAM constellation grid between $-(M-1)$ and $(M-1)$. If $\hat{x}_n^{(m+1)} \notin \{-(M-1), \dots, (M-1)\}$, then it is updated by substituting the modified $\lambda_n^{(m)}$ in (8) accordingly. The cost difference between the two consecutive LAS iterations is given as

$$\begin{aligned} \Delta \Lambda_n^{(m+1)}(\lambda_n^{(m)}) &= \Lambda^{(m+1)} - \Lambda^{(m)} \\ &= q_{nn} \left(\lambda_n^{(m)2} \right) - 2\lambda_n^{(m)} \left| z_n^{(m)} \right|. \end{aligned} \quad (10)$$

The algorithm continues until the cost difference between the two consecutive iterations is negative and is computed for all the neighbors such that

$$\mathcal{F}_0 = \arg \min_n \Delta \Lambda_n^{(m+1)}(\lambda_n^{(m)}), \forall n = 1, \dots, N_t. \quad (11)$$

If $\Delta \Lambda_n^{(m+1)}(\lambda_n^{(m)})$ is negative, then the corresponding vector update in the $(m+1)$ th iteration is given by

$$\hat{\mathbf{x}}_n^{(m+1)} = \hat{\mathbf{x}}_n^{(m)} + \lambda_n^{(m)} \text{sign}(z_n^{(m)}) \alpha_n, \quad (12)$$

$$\mathbf{z}_n^{(m+1)} = \mathbf{z}_n^{(m)} + \lambda_n^{(m)} \text{sign}(z_n^{(m)}) \beta_n, \quad (13)$$

where α_n is a unit vector with all entries equal to zero except the n th entry and β_n is the n th column of $\mathbf{Q}$ matrix. The above iteration steps are repeated until the optimum solution is obtained and the algorithm terminates if the cost difference in the $(m+1)$ th iteration is positive, i.e., $\Delta \Lambda_n^{(m+1)}(\lambda_n^{(m)}) > 0$. The main steps followed by the LAS algorithm in the estimation of $\mathcal{F}_0$ are summarized in Algorithm 1. In order to find the counter estimate required for the calculation of the soft output, we proceed to the mod-LAS algorithm-based counter

¹This work considers the less complex variant of the LAS algorithm and focuses on the efficient estimation of soft output for 1-LAS algorithm and is straightforward applicable to the M-LAS algorithm as well.

Algorithm 1 Step 1 of soft-output LAS

Data: $\hat{\mathbf{x}}^{(0)}$, $\mathbf{H}$, $\mathbf{y}$
Result: LLR matrix ($\mathbf{L}$) for $\hat{\mathbf{x}}_0$
 $\mathbf{Q} = \mathbf{H}^H \mathbf{H}$; $\mathbf{z}^{(0)} = (\mathbf{y}^{(0)} - \mathbf{H}\hat{\mathbf{x}}^{(0)}) \mathbf{H}^H$; $\hat{\mathbf{x}}_n^{(m)} = \hat{\mathbf{x}}^{(0)}$
for $n = 1 : N_t$ **do**
 while $flag = 0$ **do**
 $\hat{\mathbf{x}}_n^{(m+1)}, \mathbf{z}_n^{(m+1)} \leftarrow$ Update $\hat{\mathbf{x}}_0$ and $\mathbf{z}_n$ according to (12)
 and (13), respectively.
 $\Delta\Lambda_n^{(m+1)}(\lambda_n^{(m)}) \leftarrow$ Find the cost difference in (10).
 if $\Delta\Lambda_n^{(m+1)}(\lambda_n^{(m)}) < 0$ **then**
 $\hat{\mathbf{x}}_n^{(m)} = \hat{\mathbf{x}}_n^{(m+1)}$
 else
 $\hat{\mathbf{x}}_n^{(m+1)} = \hat{\mathbf{x}}_n^{(m)}$; $flag = 1$
 end
 end
end

hypothesis approximation outlined in the next step.

2) **Step 2:** For the soft-output estimation, the LLR computation in (4) requires the calculation of a bit-wise counter estimate for the detected symbol. On the M-QAM constellation grid, we define the counter symbols as those symbols that have a particular bit of interest flipped compared to the detected symbol. The best counter estimate for a particular symbol is selected from all the possible sets of counter symbols on the constellation grid. The search is started with an initial guess which is generated by manually flipping the j th bit in n th symbol i.e., if a j th bit of the n th detected symbol in step 1 is 0, it is flipped to 1 and vice versa based on the M-QAM constellations. Let $\hat{\mathbf{x}}_n(j)$ be the n th symbol in the detected vector with j th bit flipped. Then the initial counter estimate is denoted by $\hat{\mathbf{x}}'$. Then the best counter estimate for j th bit in n th symbol is searched within the neighborhood search space $\mathbb{S}'$ limited to $(M^{N_t})/2$ number of counter symbols on M-QAM constellation grid. It is worth noting that the remaining symbols in the hypothesis other than the symbol of interest are updated according to the constellation grid. This new search space contains only the counter symbols and the best solution from all the neighbors in $\mathbb{S}$ is selected iteratively based on the likelihood metric. The likelihood metric in the k th iteration is given by

$$\Lambda'^{(k)} = \hat{\mathbf{x}}'^{(k)T} \mathbf{G}'^{(k)} \hat{\mathbf{x}}'^{(k)} - 2\mathbf{y}^T \mathbf{H} \hat{\mathbf{x}}'^{(k)}, \quad (14)$$

where $\mathbf{G}'^{(k)} = \mathbf{H}^H \mathbf{H}$.

All the possible neighbor vectors are searched iteratively conditioned that the cost difference in the two consecutive search iterations is less than zero. The update rule in the $(k+1)$ th iteration is given by

$$\hat{\mathbf{x}}_n'^{(k+1)} = \hat{\mathbf{x}}_n'^{(k)} + \lambda_n'^{(k)} \text{sign}\left(z_n'^{(k)}\right) \alpha_n', \quad (15)$$

$$\mathbf{z}_n'^{(k+1)} = \mathbf{z}_n'^{(k)} + \lambda_n'^{(k)} \text{sign}\left(z_n'^{(k)}\right) \alpha_n'. \quad (16)$$

Algorithm 2 Step 2 of soft-output LAS and LLR computation

Data: $\hat{\mathbf{x}}, \mathbf{H}, \mathbf{y}, i, n, \mathcal{A}$
Result: LLR matrix ($\mathbf{L}$) for $\hat{\mathbf{x}}_0$
 /* Initial counter estimate with j th bit
 flipped. */
 $\hat{\mathbf{x}}'^{(k)} = \hat{\mathbf{x}}_n(j)$
for $n = 1 : N_t$ **do**
 $flag = 0$; $k = 1$
 while $flag = 0$ **do**
 for j s.t. $(\hat{\mathbf{x}}_n'^{(k+1)} \in \mathbb{S})$ **do**
 $\Delta\Lambda_n'^{(k+1)}(\lambda_n^{(k)}) \leftarrow$ The likelihood cost for all the
 counter estimate neighbors.
 end
 if $\min(\Delta\Lambda_n'^{(k+1)}(\lambda_n^{(k)})) < 0$ **then**
 $\hat{\mathbf{x}}_n'^{(k+1)} = \hat{\mathbf{x}}_n'^{(k)} + \lambda_n'^{(k)} \alpha_n'$
 $k = k + 1$
 else
 $flag = 1$
 end
 end
end
 /* Computation of LLR */
for $i = 1 : N_t$ **do**
 for $j = 1 : M$ **do**
 if $\hat{\mathbf{x}}(i, j) > 0$ **then**
 $l_{ij} = \frac{1}{\sigma^2} (\mathcal{F}_0(\hat{\mathbf{x}}) - \mathcal{F}_1(\hat{\mathbf{x}}'))$
 else
 $l_{ij} = \frac{1}{\sigma^2} (\mathcal{F}_1(\hat{\mathbf{x}}') - \mathcal{F}_0(\hat{\mathbf{x}}))$
 end
 end
end

Likewise, the cost difference in $(k+1)$ th iteration is given as

$$\Delta\Lambda_n'^{(k+1)}(\lambda_n^{(k)}) = q_{nn} \left(\lambda_n'^{(k)2} \right) - 2\Re \left(\sum_{j \neq n}^{N_t} \lambda_n'^{(k)} y_n h_{nj} + \sum_{j \neq n}^{N_t} q_{nj} \hat{x}_n'^{(k)} h_{nj} \right), \quad (17)$$

where q_{nj} is the n th row and j th column of $\mathbf{Q}$ matrix, h_{nj} is the channel gain defined in (2).

The likelihood cost must decrease from k th iteration to $(k+1)$ th iteration. The process from (15) to (17) continues and the cost difference is computed for all neighbors with $\mathbf{x}' \in \mathbb{S}'$ in an iterative fashion. The algorithm terminates when there is no decrease in the likelihood cost and the minima are achieved. The final counter estimate $\hat{\mathbf{x}}_n'^{(k+1)}$ is selected based on

$$\mathcal{F}_1 = \arg \min_n \Delta' \Lambda_n'^{(k+1)} \left(\lambda_n^{(k)} \right). \quad (18)$$

The main steps followed in the computation of the best counter estimate ($\hat{\mathbf{x}}_n'^{(k)}$) for $\hat{\mathbf{x}}_n^{(k)}$ are summarized in the first part of Algorithm 2. Once the LAS detected output and its corresponding counter estimate are obtained by solving $\mathcal{F}_0$ and $\mathcal{F}_1$, respectively, then the LLR values are computed. The steps followed in the computation for LLR values for $N_t \times 1$

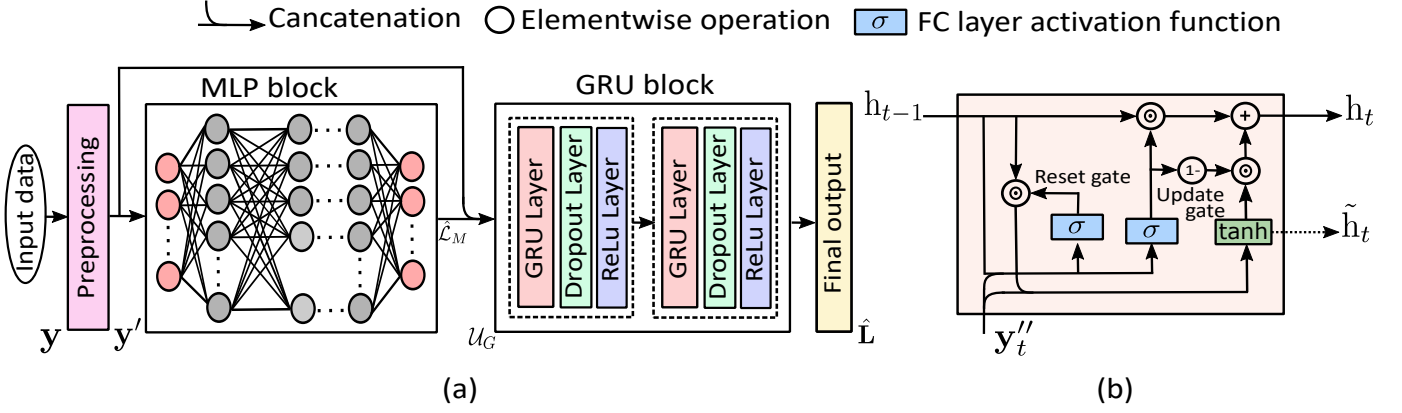


Fig. 2: DNN architecture for the proposed Deep LAS: (a) MLP-GRU-based LLR approximation (b) illustration of GRU layer's unit cell operation.

symbol vector are included in the second part of Algorithm 2.

The proposed two-step LAS algorithm efficiently estimates LLR values through the efficient computation of the detected output and its counter estimate. However, due to the strong learning capability of ML, data-driven detection techniques are more suitable in this regard, as they use a trained DNN architecture for reconstructing the received symbols.

IV. PROPOSED DEEP LAS DETECTION

In this section, we present the input and output data format for training the proposed Deep LAS followed by the Deep LAS to estimate the LLR values from the pre-processed received symbol vectors. The proposed Deep LAS is a trainable DNN framework designed to learn the soft output from received data. The Deep LAS is first trained offline with the available equalized data in the database and the trained model is then used for online estimation. The proposed Deep LAS estimates the LLR value from the received equalized complex symbol vector. For elaboration, the received symbol vector ($\mathbf{y}$), the symbol vector of MMSE estimator ($\hat{\mathbf{x}}_m$) for M-QAM symbols, and the LLR values ($\hat{\mathbf{L}}$) are represented as

$$\hat{\mathbf{y}} = \begin{bmatrix} \text{vec}(\Re(\mathbf{y})) \\ \text{vec}(\Im(\mathbf{y})) \end{bmatrix}, \hat{\mathbf{x}}_m = \begin{bmatrix} \text{vec}(\Re(\hat{\mathbf{x}}_m)) \\ \text{vec}(\Im(\hat{\mathbf{x}}_m)) \end{bmatrix},$$

$$\hat{\mathbf{L}} = [\mathbf{l}_{i1}, \mathbf{l}_{i2}, \dots, \mathbf{l}_{iB}], \quad (19)$$

where $\hat{\mathbf{y}}$ and $\hat{\mathbf{x}}_m$ are $(2N_t \times 1)$ length real equivalent vectors, respectively, while $\hat{\mathbf{L}}$ is of length $(N_t \mathbb{B} \times 1)$ where $\mathbf{l}_{ij} \forall i \in \{1, \dots, N_t\} \forall j \in \{1, \dots, \mathbb{B}\}$ denotes the LLR value for the j th bit in the i th received symbol. To estimate $\hat{\mathbf{L}}$ from $\hat{\mathbf{x}}_m$, the proposed DNN architecture for the Deep LAS is presented next.

A. MLP-GRU based network architecture

The proposed DNN for Deep LAS is inspired by the hybrid ML models with multiple learning blocks which have better learning capability compared to their counterpart standalone ML algorithms [52], [53]. The Deep LAS comprises two DNN blocks as shown in Fig. 2. The first DNN block consists of MLP and the second block includes GRU layers architecture.

The MLP block roughly estimates the LLR values which are concatenated with the equalized preprocessed received symbol data and are fed into the GRU block. The GRU block sequentially learns the LLR values from the input with data sequences and the MLP estimated rough LLR sequence. Unlike the standalone models, the proposed Deep LAS model efficiently learns the LLR values by utilizing the extended sequential input data in the consecutive blocks.

The MLP block of the proposed Deep LAS architecture consists of $\mathcal{H}$ layers, i.e., input layer, output layer, and $\mathcal{H} - 2$ number of hidden layers. The received signal vector $\mathbf{y}' = \hat{\mathbf{x}}_m$ is fed into the input layer comprised of $2N_t$ number of neurons which pass it to the hidden layers each with $\mathcal{K}$ neurons for further processing. The output obtained at the k th hidden layer can be expressed as

$$\mathcal{O}_k(\mathbf{y}') = \begin{cases} \text{ReLU}(\mathbf{w}_k \mathbf{y}' + \zeta_k) & \text{if } k = 1 \\ \text{ReLU}(\mathbf{w}_k \mathbf{y}'_{k-1} + \zeta_k) & \forall k \in \{2, \dots, \mathcal{H} - 1\} \end{cases}, \quad (20)$$

where $\text{ReLU}(x) = \max(x, 0)$ denotes the activation function while w_k and ζ_k denote the weight and bias vectors to be learned in the k th layer, respectively. The output of $(\mathcal{H} - 1)$ th layer is passed through the output layer with a single neuron to generate the final output of the MLP block and is expressed as

$$\mathcal{O}'(\mathbf{y}') = \mathcal{L}_A(\mathcal{O}_{\mathcal{H}-1}(\mathbf{y}')), \quad (21)$$

where $\mathcal{L}_A$ denotes the linear activation function.

At the output of the MLP block, a rough estimate of the $N_t \mathbb{B}$ number of LLR sequences is obtained. Let $\hat{\mathbf{L}}_M$ be the MLP estimated LLR sequence, then the input data for training the succeeding DNN block is generated by concatenating the actual input data with the rough estimate of the MLP block and is given by $\mathcal{U}_G = [\mathbf{y}'; \hat{\mathbf{L}}_M]$. The new input data has an extended dimension and improves the learning capability of the succeeding DNN block. Unlike the first block, the GRU-based DNN block follows sequential learning, where the final LLR sequence is estimated from the extended data, $\mathcal{U}_G$, with $2N_t + 1$ sequences each of $1 \times \mathcal{N}$ length. The GRU block consists of two sub-blocks and each sub-block consists of

Algorithm 3 MLP block training

Input: Training samples $\{\hat{\mathbf{x}}_m, \hat{\mathbf{L}}\}$
Output: Train network parameters Θ
 $\hat{\mathbf{x}}_m = \hat{\mathbf{x}}_m / \max(\hat{\mathbf{x}}_m) \leftarrow$ Normalize the data.

 $\mathcal{M} = \infty$
for $i = 1 : \mathcal{T}$ **do**
 $\Theta \leftarrow$ Randomize initial parameters.

 $\text{net}_i \leftarrow$ Train the network to minimize the metric (24).

if $\mathcal{M} > \min(\hat{F}_{\text{mse}}(\Theta_i))$ **then**
 $\text{net} = \text{net}_i$; $\mathcal{M} = \min(\hat{F}_{\text{mse}}(\Theta_i))$
end
end

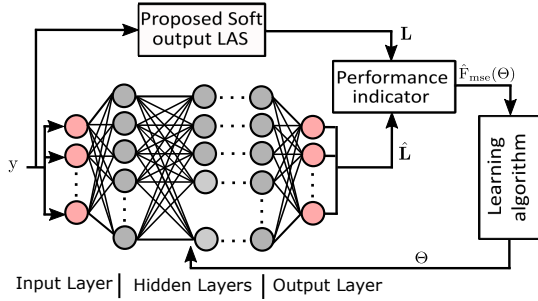


Fig. 3: Schematic of training for MLP block.

three layers, i.e., GRU, dropout, and activation layer, where the dropout layer is sandwiched between the GRU and the ReLu activation layer as shown in Fig. 2 (a). The motivation behind the use of GRU is that out of the recurrent neural network (RNN) family GRU is regarded as a modified version of LSTM and has less complexity compared to LSTM and Bi-LSTM because of fewer gates and parameters involved [54], [55]. The complexity order of BiLSTM, LSTM, and GRU is $\mathcal{C}_{\text{Bi-LSTM}} > \mathcal{C}_{\text{LSTM}} > \mathcal{C}_{\text{GRU}}$, where BiLSTM has higher complexity than LSTM as it uses multiple LSTM blocks in bi-directional pattern. The GRU layer consists of $\mathcal{V}$ number of GRU cells with an update and reset gates as shown in Fig. 2 (b). Let us denote the input and output of the $\mathcal{V}$ th GRU cell at a current timestamp t by $\mathbf{y}_t''$ and $\mathbf{h}_t$, respectively, then the unit cell transition of $\mathcal{V}$ th GRU cell is given by

$$\begin{aligned} \mathbf{h}_t &= \Psi_t \odot \mathbf{h}_{t-1} + (1 - \Psi_t) \odot \tilde{\mathbf{h}}_t, \\ \tilde{\mathbf{h}}_t &= \tanh(\mathcal{W}^c \mathbf{y}_t'' + \mathcal{X}^c (\Upsilon_t \odot \mathbf{h}_{t-1}) + \zeta^h), \\ \Upsilon_t &= \sigma(\mathcal{W}^\Upsilon \mathbf{y}_t'' + \mathcal{X}^\Upsilon \mathbf{h}_{t-1} + \zeta^\Upsilon), \\ \Psi_t &= \sigma(\mathcal{W}^\Psi \mathbf{y}_t'' + \mathcal{X}^\Psi \mathbf{h}_{t-1} + \zeta^\Psi), \end{aligned} \quad (22)$$

where $\odot$ represents the element-wise multiplication, $\mathbf{h}_{t-1}$ denotes the output at a previous timestamp, ζ^Υ , ζ^Ψ , and ζ^h are the bias vectors, $\mathcal{W}^\Upsilon$, $\mathcal{W}^\Psi$, $\mathcal{W}^c$, $\mathcal{X}^\Upsilon$, $\mathcal{X}^\Psi$, and $\mathcal{X}^c$ are the weight matrices learned during training, $\sigma(a) = 1/(1 + \exp(-a))$ and $\tanh(a) = (\exp(a) - \exp(-a)) / (\exp(a) + \exp(-a))$ denote sigmoid and tangent hyperbolic function, respectively. Considering the proposed learning architecture, the final LLR estimates by the

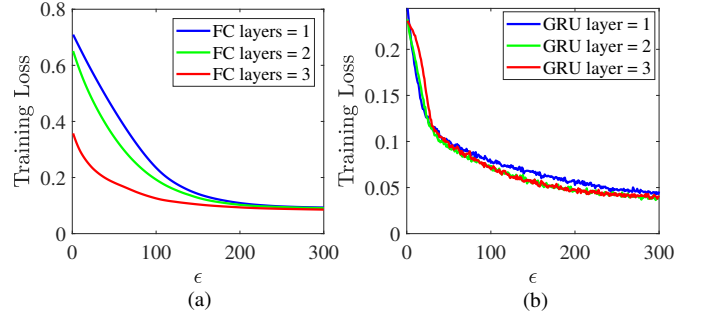
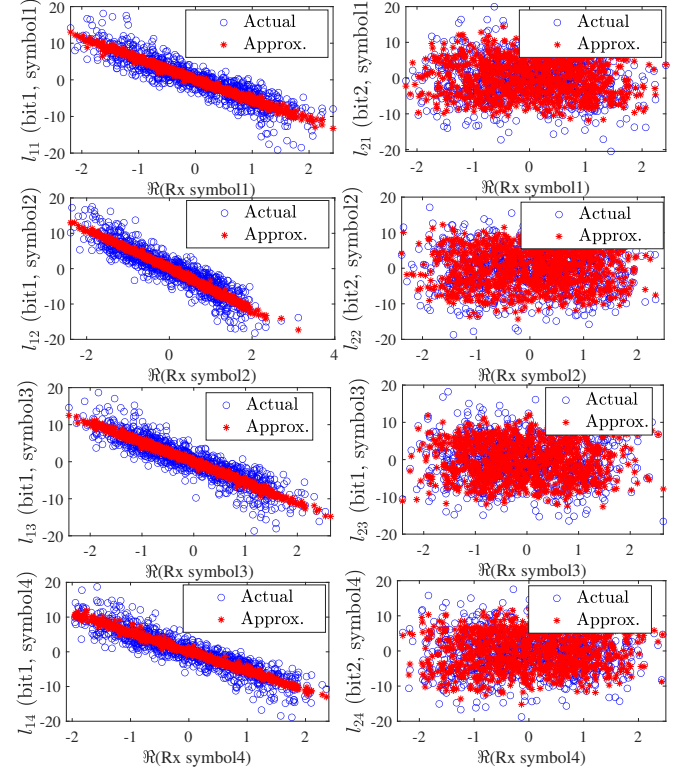


Fig. 4: Training loss of (a) MLP and (b) GRU block.


 Fig. 5: LLR approximation by MLP block for $N_t = N_r = 4$, 4-QAM modulation scheme, and SNR = 4 dB.

proposed Deep LAS can be functionally expressed as

$$\hat{\mathbf{L}} = \mathcal{O}'' \left(\left[\mathcal{O}'(\mathbf{y}'), \hat{\mathcal{L}}_M \right] \right), \quad (23)$$

where $\mathcal{O}''(\cdot)$, $\mathcal{O}'(\cdot)$, and $\hat{\mathcal{L}}_M$ denote the MLP block, the GRU block, and the rough LLR estimate provided by the MLP block, respectively.

Each block of the proposed Deep LAS is trained independently. The MLP block is trained for the range of SNR values using the pre-processed received symbol data and the target LLR values generated by the soft-output LAS algorithm. Let us denote all the trainable hyper-parameters of the MLP block by $\Theta = \{\mathbf{w}_k, \zeta_k\}$. Levenberg-Marquardt's back-propagation algorithm is used to train Θ where the trainable parameter values are randomly initialized for a fixed batch size $\mathcal{B} = 40$ in the first epoch. The training is performed for a fixed number of epochs $\epsilon = 500$ until there is no significant decrease in the

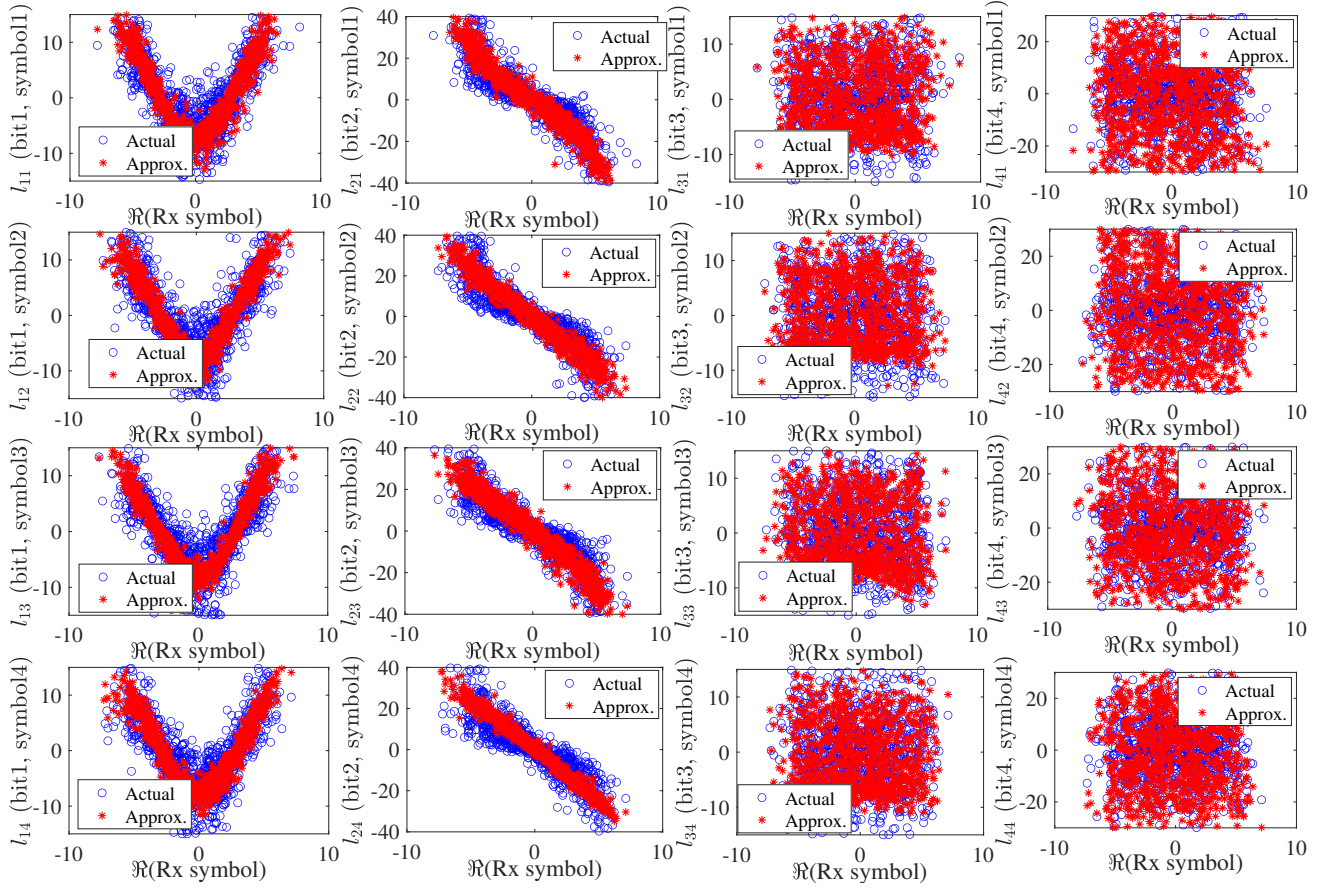


Fig. 6: LLR approximation by MLP block with $N_t = N_r = 4$, 16-QAM modulation scheme and SNR = 10 dB.

loss function beyond the pre-defined tolerance level. The root mean squared error (RMSE) function is used to evaluate the training and is given by

$$\hat{F}_{\text{rmse}}(\Theta) = \sqrt{\frac{1}{B} \sum_{i=1}^B (\hat{\mathbf{L}}_i - \mathbf{L}_i)^2}, \quad (24)$$

where $\hat{\mathbf{L}}_i$ and $\mathbf{L}_i$ denote the approximated and actual LLR vectors. The training is performed layer-wise, where $2N_t$ number of pre-processed received symbols sequences are fed to the network. For each sequence, the MLP network is trained $\mathcal{T}$ times, and one with the best performance indicator is used for online approximation of LLR values in the first block of the proposed Deep LAS. The step followed in the overall training of the MLP block is summarized in algorithm 3, whereas the schematic for each training is given in Fig. 3. The weights and biases are updated until the stopping criteria are satisfied based on the steepest descent method. The updated weights and biases are given as follows.

$$w_{k+1} = w_k - \mathcal{R} \frac{\partial \hat{F}_{\text{rmse}}(\Theta)}{\partial w}, \quad (25)$$

$$\zeta_{k+1} = \zeta_k - \mathcal{R} \frac{\partial \hat{F}_{\text{rmse}}(\Theta)}{\partial \zeta}, \quad (26)$$

where k denotes the layer index and $0 < \mathcal{R} \leq 1$ denotes the learning rate.

For MLP block, the number of neurons in the hidden layers is $\mathcal{K} = 10$ while the output layer has a single neuron. Each layer in the LLR matrix is learned from the input pre-processed received equalized data. The number of training epochs $\epsilon = 300$, $\mathcal{T} = 3$, and the value of $\mathcal{R}$ is set to 0.001. Similarly, ADAM solver is used for the GRU block and the maximum number of epochs is set to 40 while the batch size of $\mathcal{B} = 40$ is used. The GRU layer in each sub-block has $\mathcal{V} = 100$ cells and the dropout layer has dropout probability equals 0.01. During the training of each block, the randomly selected 80% of the total data points per estimation is used for training while the remaining 20% is used for validation. The training performance of the MLP and GRU block for different numbers of fully connected and GRU layers is given in Fig. 4 (a) and (b), respectively. It can be observed from Fig 4 that decreasing the number of fully connected layers in the MLP block results in performance deterioration while two GRU layers provide good training performance.

V. SIMULATION RESULTS

In this section, we evaluate the performance of the proposed model-based two-step soft-output LAS and the data-driven MLP-GRU-based Deep LAS in MIMO-OFDM systems. We first evaluated the accuracy of the LLR approximation by the proposed data-driven Deep LAS architecture compared with the actual LLR values. Then, we present the BER performance

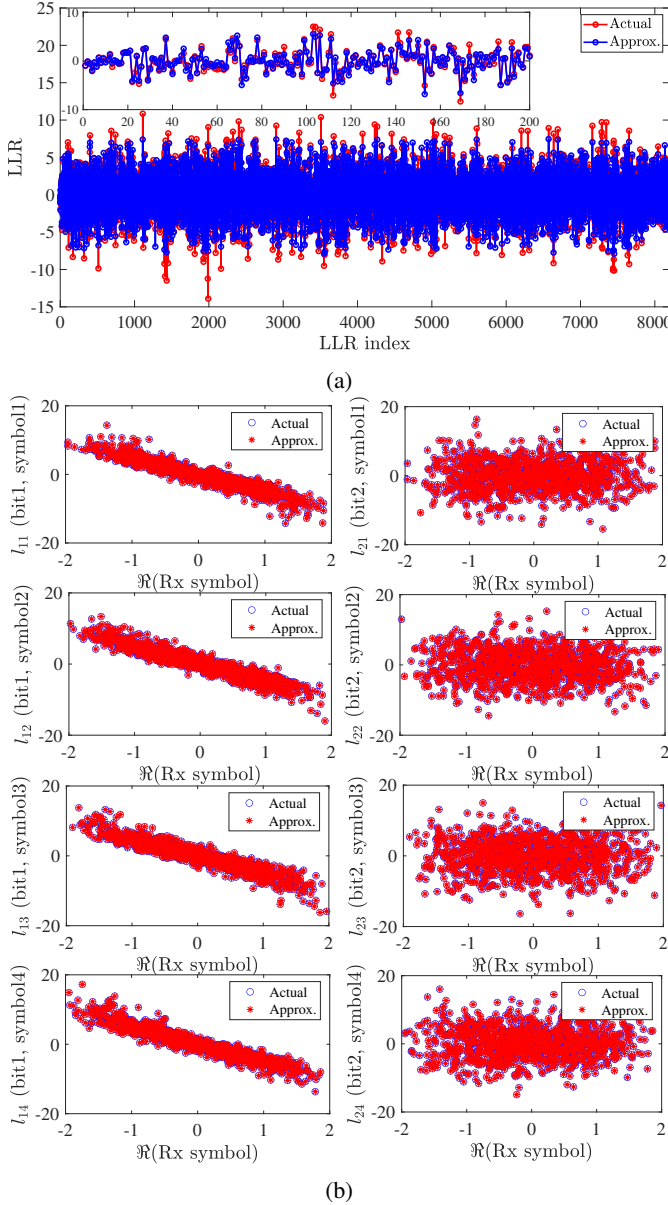


Fig. 7: LLR approximation by GRU block of the proposed Deep LAS for $N_t = N_r = 4$, 4-QAM modulation scheme, and (a) SNR = 0 dB (b) SNR = 4 dB.

of the proposed Deep LAS compared to the existing model-based and data-driven detection schemes.

In the simulation setup, we consider a MIMO-OFDM transceiver with $N_t = N_r = 4$ antenna elements, where the transmission occurs in blocks of eight symbols per channel. The IFFT and FFT lengths are set to 1024 unless stated otherwise. The decoder uses turbo codes with rate = 1/2 and eight internal iterations. M-QAM modulation with $M = 4$ and 16 is considered and random interleaver and de-interleaver are used at the transmitter and the receiver. Considering the system model in Section II, the channel between the transmitter and receiver is assumed to be Rayleigh fading. It is worth noting that the proposed data-driven Deep LAS is trained and validated with the data generated using the two-step soft-

output LAS presented in Section III. Furthermore, all the simulations are performed in Matlab R2023a installed on a 64-bit operating system with 11th Gen Intel(R) Core(TM) i7-11700K processor and 32GB RAM.

A. Block-wise LLR approximation by the proposed MLP-GRU based soft-output Deep LAS

Fig. 5 presents the LLR values approximated by the initial MLP block in comparison with the actual LLR values for the 4-QAM modulation scheme with two bits per received symbol and SNR equals 4 dB. Similarly, Fig. 6 presents the rough approximation of LLR values by the MLP block in comparison with the actual LLR values for 16-QAM modulation with 4 bits per received symbol and SNR equals 10 dB. It can be seen in Fig. 5 and Fig. 6 that the MLP provides a reasonable approximation to the LLR values for all received bits in the case of 4-QAM and 16-QAM symbols by explicitly demapping the pre-processed received equalized symbols into the corresponding LLR values. Each bit has different reliability information and variation is observed in the LLR values due to the independent generation of data bits in the modulated symbols.

Fig. 7 presents the final LLR values approximated by the second block of the proposed MLP-GRU-based Deep LAS. Fig. 7 (a) shows the final LLR approximation of the GRU block as a function of the LLR index in comparison with the actual LLR values calculated via the model-based soft-output LAS for SNR = 0 dB. Similarly, Fig. 7 (b) shows the bit-wise LLR approximation versus the real part of the received symbols in comparison with the actual LLR values for SNR = 4 dB and 4-QAM modulation scheme. It is depicted that for both SNR values, the GRU block accurately learns the target LLR values by refining the rough approximation of the MLP block using the extended data input.

Fig. 8 presents the LLR approximation of the proposed Deep LAS for 16-QAM modulation. Fig. 8 (a) shows the approximated LLR and the actual LLR values as a function of LLR index for SNR = 7 dB, while Fig. 8 (b) shows the bit-wise LLR values versus the real part of the input data for SNR = 10 dB. Fig. 8 (a) and Fig. 8 (b) depict that in the case of 16-QAM modulation scheme, the proposed MLP-GRU based Deep LAS accurately approximate the LLR for both lower and higher SNR values, respectively.

B. BER comparison of the proposed two-step LAS and Deep LAS with other model-based techniques

Fig. 9 presents the BER performance of the proposed two-stage soft-output LAS and the proposed Deep LAS in comparison with the conventional LAS algorithm. It is evident from the fact that the proposed two-step LAS achieves superior performance compared to the conventional hard output LAS through accurate computation of the soft output. Similarly, the block-wise BER performance of the proposed data-driven Deep LAS detector is better compared to the conventional hard output LAS. The rough LLR values approximated by the MLP block lead to improved performance and achieve the SNR gain of 1.9 dB and 1.8 dB with the conventional LAS

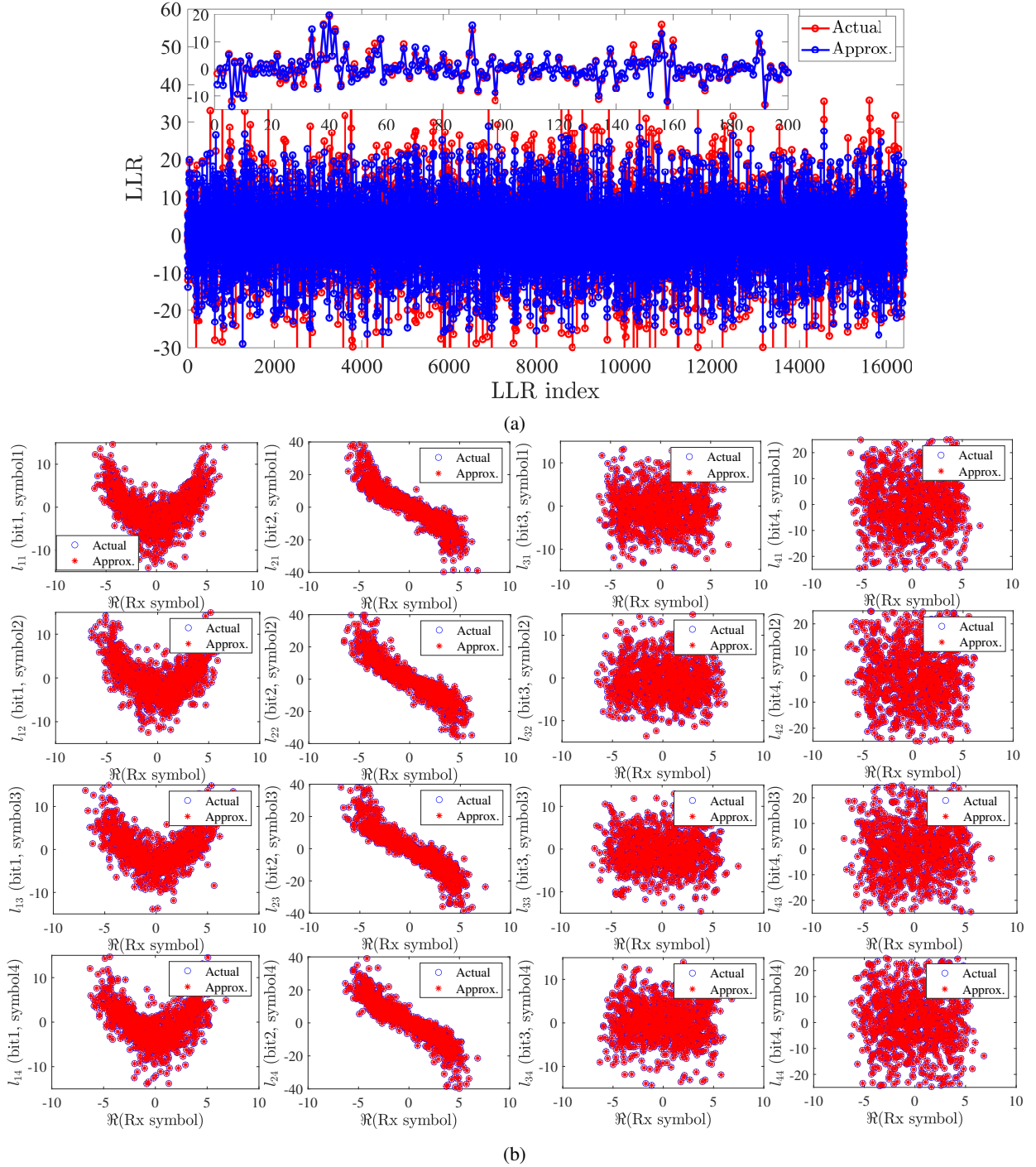


Fig. 8: LLR approximation by GRU block of the proposed Deep LAS for $N_t = N_r = 4$, 16-QAM modulation scheme and (a) SNR = 7 dB (b) SNR = 10 dB.

for 4-QAM and 16-QAM, respectively. Overall the proposed Deep LAS efficiently learns the LLR values by refining the MLP estimate through the succeeding GRU block and achieves the SNR gain of 2.55 dB and 3 dB with the conventional LAS while maintaining the BER of 10^{-4} for 4-QAM and 16-QAM modulation, respectively.

Fig. 10 provides the BER comparison of the proposed Deep LAS with the other LAS variants, well-known DetNet, and optimal SD algorithm. Fig. 10 shows that for both ZF and

MMSE as an initial filter, the proposed soft-output LAS and Deep LAS surpass the conventional LAS algorithm as well as DetNet in terms of BER for lower and higher order modulation. The proposed deep-LAS achieves better performance compared to DetNet because of the efficient estimation of soft output and relaxing the DetNet assumption of deploying more antennas at the transmitter than the receiver. Moreover, the proposed Deep LAS achieves near-optimal BER with an SNR gap of 0.4 dB and 1.2 dB with the optimal SD algorithm while

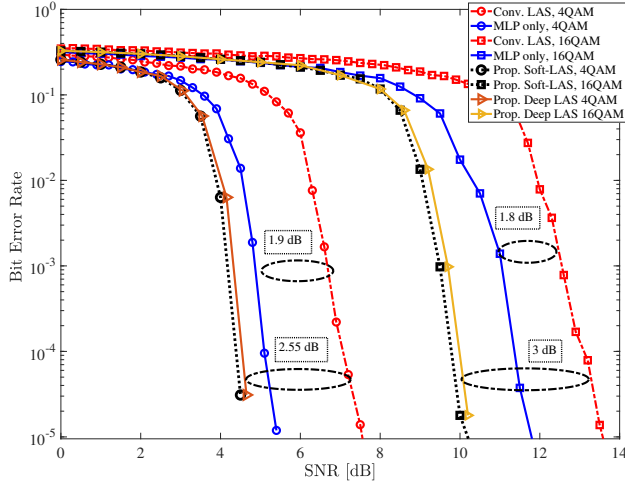


Fig. 9: BER performance of the proposed Deep LAS and the proposed two-step soft-output LAS for $N_t = N_r = 4$ and M-QAM modulation scheme.

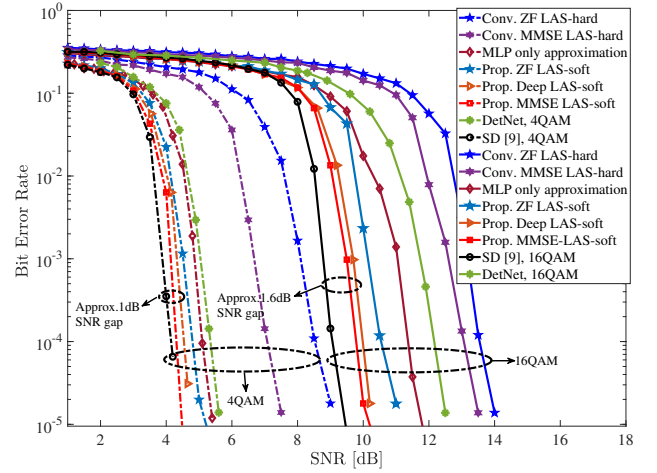


Fig. 10: BER performance of the proposed Deep LAS with other model-based detection schemes for $N_t = N_r = 4$ and M-QAM modulation scheme.

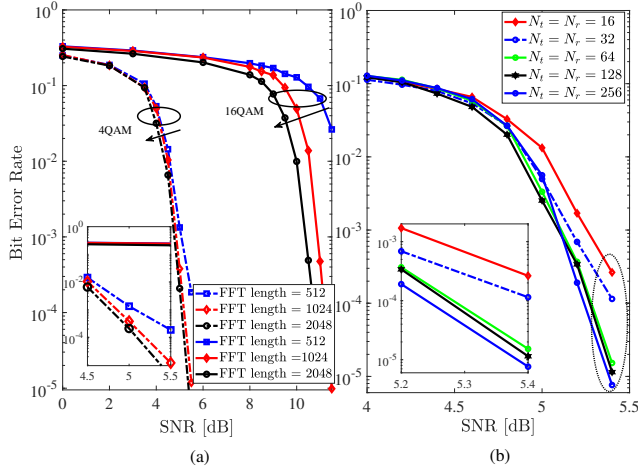


Fig. 11: BER performance of proposed detector as a function of (a) FFT length and (b) number of transmit-receive antennas.

maintaining the BER of 10^{-5} .

Fig. 11 presents the BER performance of the proposed Deep LAS for different FFT lengths and number of deployed antennas at the transmitter and receiver in the case of 4-QAM and 16-QAM, respectively. Fig. 11 (a) demonstrates that the performance of Deep LAS improves with an increase in the FFT length. For higher-order modulation, the BER performance is more affected by FFT length compared to the lower-order modulation. The reason behind this fact is that in OFDM simulations, the number of input sequences fed to the Deep LAS depends on the modulation order and the length of each sequence increases with an increase in the FFT length. Hence, leads to better training of Deep LAS in case of higher FFT length and better LLR approximation which results in improved performance compared to the lower FFT length. Similarly, Fig. 11 (b) shows the impact of the transmit-receive chain on the detection performance of the proposed soft output detector. It is observed a higher average received

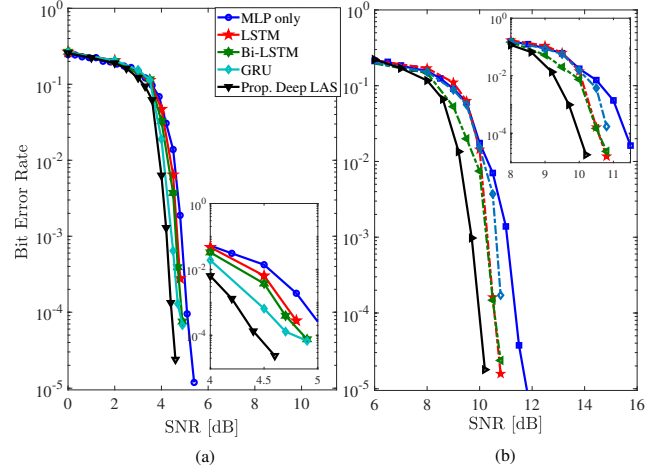


Fig. 12: Performance comparison of proposed Deep LAS with other data-driven learning techniques for (a) 4-QAM and (b) 16-QAM.

SNR is required for a MIMO system with a lower number of N_t and N_r to achieve satisfactory BER performance.

C. BER comparison of the proposed Deep LAS with data-driven learning techniques

Fig. 12 (a) and (b) provide the BER performance comparison of the proposed Deep LAS detection with the standalone MLP and other RNN-based learning techniques for 4QAM and 16QAM, respectively. The LSTM [42], Bi-LSTM [43], and GRU [44] aided LLR approximations are evaluated and compared with the proposed hybrid MLP-GRU-based Deep LAS in MIMO-OFDM systems. For the benchmark sequential learning models, we consider an LSTM/Bi-LSTM/GRU layer with 800 hidden units, a dropout layer with a dropout probability of 0.2 with ReLu activation function, and is trained for 500 epochs. Fig. 12 depicts that among the three sequential learning techniques, LSTM outperforms its counterpart, while

TABLE I: The average per symbol complexity of the proposed detection schemes in comparison with the other MIMO detectors.

Detection schemes	MLD	SD	MMSE	DetNet	Prop. LAS	Prop. Deep LAS
Average complexity	$\mathcal{O}(\sqrt{M}^{N_t} N_t^{-1})$	$\mathcal{O}(M^{N_t \beta})$	$\mathcal{O}(N_t^2)$	$\mathcal{O}(N_t^2 N_r + N_t d_z + d_v d_z)$	$\mathcal{O}(N_t^2)$	$\mathcal{O}(N_t^2 + \mathbb{B}(N_t \mathcal{H}\mathcal{K} + N_t \mathcal{V}^2 + N_t^2 \mathcal{V}))$

the proposed MLP-GRU-based Deep LAS outperforms all the stated sequential learning techniques. Furthermore, the BER performance of the standalone MLP is poor compared to the stated sequential learning benchmarks as well as the proposed Deep LAS for both 4-QAM and 16-QAM.

D. Complexity analysis

The computational complexity of a soft-output MIMO detector depends upon the computation of optimal candidate vector and soft-output estimation. It is generally computed in terms of big $\mathcal{O}$ notation which reflects the dependence of the number of floating point operations (flops) on the input size. The computational complexity of the proposed two-step soft-output LAS algorithm comprises the equalization step, the neighbor search for hypothesis, and the corresponding counter hypothesis. The per symbol computational complexity of MMSE-based initial guess is $\mathcal{O}(N_r N_t)$ [38] and the per symbol complexity for optimal candidate search in step 1 of the proposed two-step LAS is of order $\mathcal{O}(N_t^2)$. Therefore, the overall computational complexity of the proposed soft-output LAS algorithm with LLR computation in the MIMO systems with $N_t = N_r$ is of order $\mathcal{O}(N_t^2) + \mathcal{O}(N_t^2) + \mathcal{O}(\mathbb{B}N_t) \approx \mathcal{O}(N_t^2)$. Similarly, the computational complexity of MLD-based detection is $\mathcal{O}(\sqrt{M}^{N_t} N_t^{-1})$ which increases with increase in the number of antenna chains and modulation order [32]. The complexity of SD with multiple node search and pre-detection QR decomposition is $\mathcal{O}(M^{N_t \beta})$, where $\beta \in \{0, 1\}$ is a noise dependent factor [9]. The computational complexity of the proposed data-driven architecture is the sum of the computation involved in the MLP and GRU block. The complexity of the MLP block depends on the number of layers, $\mathcal{H}$, and the number of neurons per layer, $\mathcal{K}$, and is of order $\mathcal{O}(\mathcal{H}\mathcal{K})$. The complexity of the GRU layer involves the computation of intermediate variables in (22) and the complexity of each is $\mathcal{O}(F_l \mathcal{V}(3\mathcal{V} + 3\mathcal{D} + 3))$ which becomes $\mathcal{O}(\mathcal{V}(\mathcal{V} + \mathcal{D}))$ for fixed sequence length F_l , where $\mathcal{V}$ and $\mathcal{D}$ represent the number of GRU units and the dimension of the input sequence, respectively. In total, the computational complexity of the proposed architecture is $\mathcal{O}(N_t^2 + \mathbb{B}(N_t \mathcal{H}\mathcal{K} + N_t \mathcal{V}^2 + N_t^2 \mathcal{V})) \approx \mathcal{O}(\mathbb{B}N_t^2)$. Moreover, the computational complexity of the DetNet is given as $\mathcal{O}(N_t^2 N_r + N_t d_z + d_v d_z)$, where d_z and d_v represent the augment variables [49]. The comparative analysis of the proposed detection and the benchmark schemes in terms of average per symbol computational complexity is summarized in Table I. From the analysis, it is evident that the proposed detection schemes are comparable with the linear MMSE and benchmark DetNet while it is far better than the MLD and SD in terms of computational complexity.

VI. CONCLUSION

In this paper, we propose a model-based and data-driven soft-output LAS detection scheme for MIMO-OFDM systems. The proposed model-based two-step soft-output LAS follows the detection of the transmitted symbols via conventional hard output 1-LAS while the corresponding counter hypothesis for the bit-wise soft output generation is estimated using a modified LAS algorithm. The proposed two-step LAS achieves near-optimal BER performance with less computational complexity. Furthermore, we propose a data-driven soft-output Deep LAS detection scheme that utilizes a trainable DNN architecture to approximate the soft output directly from the received equalized signals. In this regard, we propose a hybrid learning framework that comprises an MLP block followed by the GRU learning block. The proposed MLP-GRU-based Deep LAS directly learns the channel behavior and predicts the layer-wise soft output information from the pre-processed received symbols data. The proposed Deep LAS with MLP-GRU architecture efficiently predicts the LLR values and outperforms the other model-based, counterpart sequential learning-based standalone data-driven models, and DetNet by achieving a BER performance nearly equal to the model-based SD detector. The performance of the proposed detection schemes can be further explored in a doubly dispersive environment and for a more realistic 3GPP channel model in future research.

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